ROUGHTOOLS.COM

Complete Project Documentation

405	15	707	22
Tools	Categories	SSG Pages	DB Models

Next.js 14	**TypeScript**	**Tailwind CSS**	**Express.js**	**PostgreSQL**	**Prisma**	**Redis**
BullMQ	**Stripe**	**JWT Auth**	**Sentry**	**Turborepo**	**pnpm**	**Docker**

Overall Completion: ~70%



70% done | 30% remaining (192 tool configs needed)

Generated: May 26, 2026 | shahidwattoo0111@gmail.com

1. Project Overview

Project	ROUGHTOOLS.COM
Repo	ai-tools-platform (monorepo)
Framework	Next.js 14 — App Router (SSG)
Language	TypeScript (strict)
Styling	Tailwind CSS JIT
Monorepo	pnpm workspaces + Turborepo
Backend	Express.js API (apps/api)
Database	PostgreSQL + Prisma ORM
Cache	Redis
Queue	BullMQ
PDF	Gotenberg (self-hosted)
Billing	Stripe
Auth	JWT (bcrypt, sessions)
Monitoring	Sentry + Google Analytics + Microsoft Clarity
Deploy	Vercel (web) + VPS/Docker (api)
Build Output	707 static pages (SSG)
Total Tools	405 tools across 15 categories
Node	18+

2. Complete Folder Structure

Monorepo Root

```
ai-tools-platform/
apps/
web/ ← Next.js 14 App Router (frontend)
api/ ← Express.js backend
packages/
cache/ ← Redis cache helpers
calculators-core/ ← Shared calculation logic
config/ ← Shared ESLint + TS config
core-logic/ ← Core business logic
db/ ← Prisma schema + migrations (22 models)
engine/ ← Tool engine
errors/ ← Shared error types
legal/ ← Legal document helpers
queue/ ← BullMQ job queue setup
```

```
redis/ ← Redis client
seo/ ← SEO helpers + JSON-LD generators
storage/ ← AWS S3 / file storage
types/ ← Shared TypeScript interfaces
ui/ ← Shared UI components
utils/ ← Shared utility functions
validation/ ← Zod validation schemas
validators/ ← Input validators
.env.example ← All env vars documented
Makefile ← 40+ dev/build/db/docker commands
Dockerfile.gotenberg
docker-compose.yml
turbo.json
pnpm-workspace.yaml
```

apps/web/src/ — Frontend Structure

```
app/
layout.tsx Root layout (Inter font, Header, Footer, Chatbot)
page.tsx HOME PAGE
globals.css Design system, animations, CSS variables
loading.tsx / not-found.tsx / error.tsx / global-error.tsx
manifest.ts / robots.ts / sitemap.ts
sitemap-calculators.ts + sitemap-converters.ts + sitemap-developer.ts
sitemap-legal-tools.ts + sitemap-network.ts + sitemap-pdf.ts
sitemap-real-estate.ts + sitemap-text.ts
opengraph-image.tsx
(auth)/
signin/page.tsx + signup/page.tsx
about/page.tsx + blog/page.tsx + blog/[slug]/page.tsx
contact/page.tsx + pricing/page.tsx + search/page.tsx
privacy-policy/page.tsx + terms-of-service/page.tsx
settings/page.tsx + profile/page.tsx + api-docs/page.tsx
link-to-us/page.tsx + embed-generator/page.tsx
dashboard/page.tsx + dashboard/api-keys/page.tsx + dashboard/referral/page.tsx
admin/analytics/page.tsx
billing/success/page.tsx
api/
ai/route.ts
api-keys/route.ts + [id]/route.ts
auth/me/ + profile/ + register/ (route.ts each)
billing/checkout/route.ts
health/route.ts
pdf/ (20 PDF processing routes)
users/route.ts + [id]/route.ts
tools/
page.tsx Browse all tools
[category]/page.tsx + [category]/[tool]/page.tsx (generic fallbacks)
pdf/[pdfTool]/page.tsx + _components/ (PDFToolClient, pdf-tool-configs.ts)
calculators/ (21 individual tool folders, each with page.tsx + Client.tsx)
developer/ (19 tool folders)
image-tools/ (38 tool folders)
```

```
text/ (10 tool folders)
converters/ (8 tool folders)
security/ (8 tool folders)
legal-tools/ (8 tool folders)
real-estate/ (8 tool folders)
network/ (7 tool folders)
business/ (6 tool folders)
environment/ (6 tool folders)
document-tools/ (42 sub-pages incl. PDF tools)
engineering/page.tsx + [slug]/page.tsx + _components/
EngineeringCalculatorClient.tsx + engineering-configs.ts
medical/page.tsx + [slug]/page.tsx + _components/
MedicalCalculatorClient.tsx + medical-configs.ts
```

apps/web/src/ — Components & Lib

```
components/
SiteHeader.tsx Sticky header + 15-category slide-in drawer
Footer.tsx Brand + categories + popular tools + company
CategoryGrid.tsx Home bento grid (4 featured + 11 compact)
CategoryHubLayout.tsx Shared server layout for category hubs
SearchBar.tsx / ToolCard.tsx / RelatedTools.tsx / Breadcrumbs.tsx
Chatbot.tsx (lazy) + CookieConsent.tsx (lazy)
JsonLd.tsx + LazySection.tsx + OptimizedImage.tsx
ToolAdUnit.tsx (AdSense) + ThemeProvider.tsx + icons.tsx
analytics/ AdSense.tsx + GoogleAnalytics.tsx + MicrosoftClarity.tsx
SearchConsoleVerification.tsx + WebVitals.tsx
layout/ Header.tsx + Footer.tsx + Sidebar.tsx (+ .module.css each)
sections/ Hero.tsx + StatsBar.tsx + Ticker.tsx + CalculatorsSection.tsx
DocumentsSection.tsx + FilterBar.tsx + RoadmapSection.tsx
tools/ AgeCalculator + BMICalculator + CompoundInterest + DaysBetweenDates
MortgageCalculator + PercentageCalculator + RoiCalculator
TaxCalculator + TipCalculator
ui/ CategoryPage.tsx + CustomCursor.tsx + ToolCard.tsx

lib/
tools-registry.ts SOURCE OF TRUTH — all 415 tool definitions
seo.ts + seo-config.ts + structured-data.ts + og-utils.ts
relatedTools.ts + social-config.ts + canonical.ts
tool-metadata.ts + blog-posts.ts + analytics.ts + prisma.ts + utils.ts

hooks/ useAuth.ts + usePageTracking.tsx + useTheme.ts
middleware.ts + instrumentation.ts
```

apps/api/src/ — Backend Structure

```
index.ts Main Express server entry point

lib/
gotenberg.ts Gotenberg PDF service client
logger.ts + prisma.ts + sentry.ts

middleware/
csrfProtection.ts + errorHandler.ts + fileValidator.ts
notFoundHandler.ts + pdfErrorHandler.ts + pdfRateLimit.ts
```

```
rateLimiter.ts + requestLogger.ts + requireAuth.ts
sanitizer.ts + security-headers.ts + validate.ts + validateBody.ts

routes/
auth.ts + billing.ts + blog.ts + contact.ts + converters.ts
css-minifier.ts + developer.ts + gdpr.ts + health.ts
ip-checker.ts + js-beautifier.ts + metadata-remover.ts
network.ts + newsletter.ts + openapi-spec.ts + public-api.ts
referral.ts + ssl-checker.ts + temp-email.ts + text.ts
tool-analytics.ts + tool-requests.ts + url-shortener.ts + vitals.ts
affiliate.ts + api-keys.ts

calculators/ bmi.ts + calorie.ts + currency.ts + loan.ts
mortgage.ts + roi.ts + tax.ts + unit-converter.ts

pdf/ index.ts + merge.ts + split.ts + compress.ts + rotate.ts
crop.ts + protect.ts + unlock.ts + sign.ts + watermark.ts
ocr.ts + repair.ts + redact.ts + extract-pages.ts
remove-pages.ts + page-numbers.ts + organize.ts
compare.ts + pdfa.ts + convert-to-pdf.ts + convert-from-pdf.ts

developer/ js-beautifier.route.ts

utils/ pagination.ts + response.ts
jobs/ cleanupFiles.ts
```

3. Database Schema (22 Models)

PostgreSQL database managed with Prisma ORM. Schema located at packages/db/prisma/schema.prisma.

Model	Description
User	Core user account: email, username, passwordHash, role, emailVerified, isActive, avatar, timestamps
Session	JWT session tokens with userId, token (unique), userAgent, ipAddress, expiresAt
Category	Tool categories: slug (unique), name, sortOrder
Tag	Tags for tools and blog posts: slug, name, color
Tool	Tool entries: slug, name, description, categoryId, status (DRAFT/PUBLISHED/ARCHIVED)
ToolTag	Many-to-many join table between Tool and Tag
Comment	User comments on tools: content, userId, toolId, timestamps
Favorite	User favorites: userId + toolId (unique together)
BlogPost	Blog articles: title, slug, content, status, authorId, publishedAt
ApiUsageLog	API usage tracking: endpoint, method, statusCode, userId, ipHash, responseTime
ToolUsage	Tool usage analytics: toolSlug, userId, ipHash, sessionId, timestamps
WebVital	Core Web Vitals: name (LCP/FID/CLS/TTFB), value, page, deviceType
ToolRequest	User-submitted tool requests: title, description, votes, status
NewsletterSubscriber	Email newsletter: email (unique), confirmedAt, source
ContactMessage	Contact form submissions: name, email, subject, message, status
Subscription	Stripe subscriptions: userId, stripeCustomerId, stripePriceId, status, plan
AffiliateClick	Affiliate tracking clicks: affiliateId, toolSlug, ipHash, referrer
AffiliateConversion	Affiliate conversions: affiliateId, userId, revenue, commissionRate
Referral	Referral program: referrerId, referredId, status, rewardAmount
ApiKey	Developer API keys: userId, keyHash, name, permissions, lastUsedAt
ReferralCode	Referral codes: code (unique), userId, usageCount, maxUsage
ShortLink	URL shortener: code (unique), originalUrl, userId, clickCount, expiresAt

4. All 405 Tools by Category

#	Category	Tools	Route
1	Engineering	126	`/tools/engineering`
2	Medical & Health	113	`/tools/medical`
3	Image Tools	38	`/tools/image-tools`
4	PDF Tools	26	`/tools/pdf`
5	Calculators	21	`/tools/calculators`
6	Developer Tools	19	`/tools/developer`
7	Text Tools	10	`/tools/text`
8	Document Tools	9	`/tools/document-tools`
9	Security	8	`/tools/security`
10	Unit Converters	8	`/tools/converters`
11	Legal Tools	8	`/tools/legal-tools`
12	Real Estate	8	`/tools/real-estate`
13	Network Tools	7	`/tools/network`
14	Business Tools	6	`/tools/business`
15	Environment	6	`/tools/environment`
	TOTAL	405	

PDF Tools (26 tools) /tools/pdf/

1. compress-pdf	2. merge-pdf	3. split-pdf
4. pdf-to-word	5. pdf-to-excel	6. pdf-to-powerpoint
7. pdf-to-jpg	8. pdf-to-html	9. word-to-pdf
10. excel-to-pdf	11. jpg-to-pdf	12. protect-pdf
13. unlock-pdf	14. rotate-pdf	15. crop-pdf
16. add-watermark	17. ocr-pdf	18. repair-pdf
19. redact-pdf	20. add-page-numbers	21. organize-pdf
22. extract-pdf-pages	23. remove-pdf-pages	24. compare-pdf
25. pdf-to-pdfa	26. sign-pdf	

Calculators (21 tools) /tools/calculators/

1. bmi-calculator	2. loan-calculator	3. mortgage-calculator
4. compound-interest-calculator	5. age-calculator	6. percentage-calculator
7. date-difference-calculator	8. tip-calculator	9. tax-calculator
10. currency-converter	11. gpa-calculator	12. calorie-calculator
13. scientific-calculator	14. discount-calculator	15. roi-calculator

16. inflation-calculator	17. retirement-calculator	18. salary-calculator
19. body-fat-calculator	20. ovulation-calculator	21. due-date-calculator

Medical & Health (113 tools) /tools/medical/

Cardiology & Cardiovascular (12)

1. cha2ds2-vasc-score	2. heart-score-calculator	3. framingham-risk-score
4. mean-arterial-pressure-calculator	5. cardiac-output-fick-calculator	6. timi-risk-score
7. qtc-interval-calculator	8. ldl-cholesterol-calculator	9. warfarin-dose-estimator
10. has-bled-bleeding-score	11. blood-pressure-classification	12. grace-acs-score

Nephrology & Renal (10)

1. creatinine-clearance-cockcroft-gault	2. egfr-ckd-epi-calculator	3. mdrd-egfr-calculator
4. urine-anion-gap-calculator	5. fena-feurea-calculator	6. dialysis-adequacy-ktv
7. calcium-correction-calculator	8. renal-drug-dose-adjuster	9. water-deficit-hypernatremia
10. kidney-failure-risk-equation		

Critical Care & Emergency (11)

1. curb-65-pneumonia-score	2. glasgow-coma-scale	3. sofa-sepsis-score
4. abg-interpreter	5. apache-ii-score	6. anion-gap-calculator
7. wells-score-dvt-pe	8. iv-fluid-rate-calculator	9. qsofa-sepsis-score
10. burn-surface-area-rule-of-9s	11. pecarn-head-injury-tool	

Psychiatry & Neurology (9)

1. phq9-depression-screen	2. gad7-anxiety-score	3. nihss-stroke-scale
4. epworth-sleepiness-scale	5. mmse-cognitive-screen	6. lithium-dose-calculator
7. audit-c-alcohol-screening	8. ciwa-ar-withdrawal-scale	9. ptsd-pcl5-checklist

Endocrinology & Metabolism (8)

1. visual-acuity-converter	2. hba1c-glucose-converter	3. bmi-calculator
4. insulin-dose-calculator	5. bmr-tdee-calculator	6. findrisc-diabetes-risk
7. steroid-dose-converter	8. thyroid-function-interpreter	

Pharmacology & Drug Dosing (13)

1. ideal-body-weight-calculator	2. body-fat-percentage-calculator	3. carb-glycemic-calculator
4. drug-interaction-checker	5. weight-based-drug-dose-calculator	6. drug-half-life-calculator
7. opioid-equianalgesic-converter	8. vancomycin-dosing-calculator	9. aminoglycoside-dose-calculator
10. iv-drip-rate-mcg-kg-min	11. pediatric-emergency-drug-calculator	12. pill-identifier-tool
13. medication-cost-estimator		

Obstetrics & Gynecology (10)

1. due-date-edd-calculator	2. gestational-age-calculator	3. fetal-weight-estimator

4. pregnancy-weight-gain-calculator	5. ivf-success-rate-calculator	6. ovulation-fertile-window-calculator
7. bishop-score-calculator	8. preeclampsia-risk-calculator	9. postpartum-hemorrhage-calculator
10. apgar-score-calculator		

Pulmonology & Respiratory (6)

1. spirometry-fev1-fvc-interpreter	2. psi-port-pneumonia-score	3. alveolar-arterial-gradient
4. pao2-fio2-ratio-ards	5. peak-flow-calculator	6. pack-year-calculator

Gastroenterology & Hepatology (6)

1. meld-score-calculator	2. child-pugh-score-calculator	3. saag-calculator
4. ransons-criteria-pancreatitis	5. h-pylori-eradication-tracker	6. bowel-prep-dosing-calculator

Oncology & Hematology (8)

1. chemotherapy-dose-bsa-calculator	2. anc-absolute-neutrophil-count	3. psa-velocity-doubling-time
4. tnm-staging-interpreter	5. inr-pt-ratio-converter	6. ipss-mds-score
7. blood-transfusion-volume-calculator	8. cancer-survival-probability	

Pediatrics & Neonatology (8)

1. child-growth-percentile	2. pediatric-bmi-for-age	3. fever-dose-calculator
4. holliday-segar-maintenance-fluid	5. vaccine-schedule-checker	6. corrected-gestational-age-calculator
7. pediatric-ett-size-depth	8. neonatal-jaundice-bhutani	

General Health & Wellness (12)

1. bmi-health-calculator	2. calorie-deficit-calculator	3. heart-rate-zone-calculator
4. daily-water-intake-calculator	5. sleep-cycle-calculator	6. health-insurance-premium-estimator
7. blood-pressure-tracker-chart	8. cholesterol-ratio-calculator	9. surgery-cost-estimator
10. drug-rehab-cost-calculator	11. therapy-session-cost-estimator	12. medication-cost-comparator

Engineering (126 tools) /tools/engineering/

Civil & Structural Engineering (18)

1. beam-deflection-calculator	2. concrete-mix-design	3. column-load-capacity
4. retaining-wall-stability	5. foundation-bearing-capacity	6. steel-section-properties
7. slope-stability-fos	8. rebar-quantity-estimator	9. concrete-volume-calculator
10. wind-load-calculator	11. soil-bearing-pressure	12. truss-member-force
13. moment-of-inertia-calculator	14. asphalt-pavement-thickness	15. earthwork-cut-fill
16. pile-load-capacity	17. slab-reinforcement-designer	18. construction-cost-estimator

Electrical Engineering (18)

1. ohms-law-calculator	2. wire-cable-size	3. voltage-drop-calculator
4. three-phase-power-calculator	5. transformer-sizing	6. motor-starting-current
7. solar-panel-battery-sizing	8. led-energy-savings	9. circuit-breaker-sizing
10. capacitor-bank-sizing	11. ups-battery-backup	12. power-factor-correction
13. short-circuit-current	14. conduit-fill-calculator	15. load-schedule-panel
16. pv-system-roi	17. battery-state-of-charge	18. earthing-grounding-resistance

Mechanical Engineering (18)

1. bolt-torque-calculator	2. gear-ratio-calculator	3. hp-kw-converter
4. pressure-vessel-thickness	5. fluid-flow-rate	6. heat-transfer-calculator
7. pump-head-flow	8. spring-constant-calculator	9. belt-pulley-speed
10. cnc-feeds-speeds	11. tolerance-stack-up	12. fatigue-life-estimator
13. thermal-expansion-calculator	14. bearing-life-calculator	15. stress-strain-calculator
16. reynolds-number-calculator	17. vibration-frequency	18. hydraulic-cylinder-force

HVAC & Building Services (10)

1. cooling-load-calculator	2. heating-load-estimator	3. duct-sizing-calculator
4. psychrometric-calculator	5. cop-eer-calculator	6. u-value-r-value
7. ventilation-rate-calculator	8. chilled-water-pipe-sizing	9. boiler-efficiency-calculator
10. building-energy-audit		

Chemical & Process Engineering (10)

1. pipe-pressure-drop	2. orifice-plate-flow	3. heat-exchanger-sizing
4. gas-compressor-power	5. pump-npsh-calculator	6. flare-stack-sizing
7. distillation-column-estimator	8. gas-molecular-weight	9. steam-table-lookup
10. relief-valve-sizing		

Geotechnical Engineering (8)

1. soil-bearing-capacity	2. settlement-calculator	3. slope-stability-bishop
4. lateral-earth-pressure	5. liquefaction-potential	6. pile-driving-formula
7. permeability-seepage	8. spt-n-value-parameters	

Water Resources & Environmental (8)

1. stormwater-runoff	2. water-demand-estimator	3. water-tank-sizing
4. wastewater-flow-estimator	5. solar-irradiance-pv-yield	6. engineering-carbon-footprint
7. flood-frequency-analysis	8. water-quality-index	

Electronics & PCB Design (10)

1. resistor-color-code	2. rc-filter-designer	3. antenna-length-calculator
4. pcb-trace-width	5. timer-555-calculator	6. db-signal-loss
7. op-amp-gain-calculator	8. led-resistor-calculator	9. capacitor-charge-time
10. zener-voltage-regulator		

Aerospace & Automotive (8)

1. thrust-to-weight-ratio	2. drag-coefficient-estimator	3. wing-loading-calculator
4. vehicle-fuel-economy	5. ev-range-calculator	6. engine-horsepower-estimator
7. mach-number-converter	8. braking-distance-calculator	

IT Infrastructure & Cloud (10)

1. cloud-cost-estimator	2. server-rack-power-cooling	3. network-bandwidth-calculator
4. api-load-calculator	5. database-storage-estimator	6. cyber-security-risk
7. saas-pricing-ltv	8. cicd-pipeline-cost	9. docker-k8s-sizing
10. password-entropy-calculator		

Project Management & Cost (8)

1. critical-path-method	2. boq-material-take-off	3. project-roi-calculator
4. earned-value-management	5. labour-productivity-estimator	6. crane-lift-capacity
7. contractor-markup-calculator	8. depreciation-calculator	

Image Tools (38) /tools/image-tools/

1. image-compressor	2. background-remover	3. image-resizer
4. image-converter	5. passport-photo-maker	6. image-cropper
7. bulk-image-resizer	8. image-upscaler	9. dpi-converter
10. pixel-density-calculator	11. background-changer	12. white-background-maker
13. heic-to-jpg	14. png-to-jpg	15. jpg-to-png
16. image-to-base64	17. base64-to-image	18. image-enhancer
19. image-colorizer	20. blur-background	21. add-watermark
22. remove-watermark	23. image-flipper-rotator	24. grayscale-converter
25. add-border	26. round-corner-image	27. image-metadata-viewer
28. image-color-picker	29. dominant-color-extractor	30. image-size-checker
31. meme-generator	32. photo-to-cartoon	33. id-card-photo-maker
34. thumbnail-maker	35. photo-collage-maker	36. gif-maker
37. screenshot-to-pdf	38. image-to-text	

Developer Tools (19) /tools/developer/

1. json-formatter	2. base64-encoder	3. regex-tester
4. css-minifier	5. js-beautifier	6. uuid-generator
7. hash-generator	8. color-picker	9. url-encoder
10. html-encoder	11. jwt-decoder	12. cron-parser
13. sql-formatter	14. xml-formatter	15. yaml-validator
16. markdown-editor	17. diff-checker	18. chmod-calculator
19. timestamp-converter		

Document Tools (9) /tools/document-tools/

1. invoice-generator	2. resume-builder	3. cover-letter-builder
4. business-letter	5. meeting-minutes	6. expense-report
7. proposal-generator	8. reference-letter	9. contract-creator

Text Tools (10) /tools/text/

1. word-counter	2. case-converter	3. lorem-ipsum-generator
4. text-summarizer	5. grammar-checker	6. readability-score
7. text-to-speech	8. remove-duplicate-lines	9. text-reverser
10. find-replace		

Network Tools (7) /tools/network/

1. ip-checker	2. dns-lookup	3. whois-lookup
4. port-checker	5. ping-tool	6. http-headers-checker
7. website-speed-test		

Security Tools (8) /tools/security/

1. qr-code-generator	2. password-generator	3. url-shortener
4. temporary-email	5. file-encryptor	6. vpn-checker
7. ssl-checker	8. metadata-remover	

Unit Converters (8) /tools/converters/

1. temperature-converter	2. length-converter	3. weight-converter
4. area-converter	5. volume-converter	6. speed-converter
7. time-converter	8. digital-storage-converter	

Legal Tools (8) /tools/legal-tools/

1. privacy-policy-generator	2. terms-of-service-generator	3. nda-generator
4. lease-agreement-generator	5. employment-contract-generator	6. will-generator
7. power-of-attorney-generator	8. cease-and-desist-generator	

Real Estate (8) /tools/real-estate/

1. rent-vs-buy-calculator	2. home-affordability-calculator	3. property-tax-calculator
4. closing-costs-calculator	5. cap-rate-calculator	6. rental-yield-calculator
7. home-value-estimator	8. down-payment-calculator	

Business Tools (6) /tools/business/

1. sales-copy-ai	2. ad-creative-generator	3. social-media-scheduler
4. email-marketing-ai	5. landing-page-builder	6. seo-optimizer

Environment Tools (6) /tools/environment/

1. solar-savings-calculator	2. home-energy-audit	3. ev-carbon-offset
4. flight-carbon-calculator	5. carbon-footprint-calculator	6. waste-reduction-planner

5. Project Completion Audit

Layer	Status	Done	Notes
Infrastructure & DevOps		100%	Turborepo, pnpm, Docker, ESLint, Prettier, Husky
Database Schema		100%	22 Prisma models, migrations ready
Backend API Routes		85%	Auth, Billing, PDF, Calculators, Network done
Home Page		100%	Hero, Ticker, Categories, Features, Trust strip
Category Hub Pages (15)		100%	All 15 category pages complete
PDF Tools (26)		100%	Dynamic routing + 29 configs
Calculators (21)		100%	Individual pages + client components
Developer Tools (19)		100%	Individual pages + client components
Image Tools (38)		100%	Individual pages + client components
Text Tools (10)		100%	Individual pages + client components
Unit Converters (8)		100%	Individual pages + client components
Security Tools (8)		100%	Individual pages + client components
Legal Tools (8)		100%	Individual pages + client components
Real Estate (8)		100%	Individual pages + client components
Network Tools (7)		100%	Individual pages + client components
Business Tools (6)		100%	Individual pages + client components
Environment Tools (6)		100%	Individual pages + client components
Document Tools (9)		100%	Individual pages + templates
SEO (sitemaps, OG, JSON-LD)		100%	8 sitemaps, robots.txt, OG images
Auth / Dashboard / Profile		100%	Sign in, sign up, dashboard, settings
Analytics / Monitoring		100%	GA4, Clarity, AdSense, Sentry, WebVitals
Makefile		100%	40+ commands: dev, build, db, docker, deploy

Engineering Configs (126)		17%	22/126 done — 104 calculation functions needed
Medical Configs (113)		22%	25/113 done — 88 calculation functions needed
Stripe Billing		60%	Code done, needs real PRICE_ID env vars
Email Templates (Resend)		20%	Route exists, email templates not built
Admin Panel		10%	Only analytics page — no user/content mgmt
Blog CMS		30%	Structure ready, no content pipeline
AI Tool Backends		30%	UI done, API connections incomplete
Image Processing (AI)		25%	UI done, AI service not connected

Completed Items (DONE)

- [DONE] Full monorepo infrastructure: Turborepo, pnpm workspaces, ESLint, Prettier, Husky pre-commit hooks
- [DONE] 22-model Prisma database schema (User, Session, Tool, Blog, Analytics, Billing, Referral, ApiKey, ShortLink...)
- [DONE] Redis cache layer + BullMQ queue setup
- [DONE] Sentry error monitoring + Google Analytics + Microsoft Clarity + AdSense
- [DONE] Complete Next.js home page: hero with animated orbs, marquee ticker, category bento grid, features, popular tools, trust strip
- [DONE] All 15 category hub pages with CategoryHubLayout server component
- [DONE] SiteHeader: sticky/transparent→glass-on-scroll, hamburger drawer with 15 categories + accordion
- [DONE] Footer: brand column, categories, popular tools, company links, trust badges
- [DONE] SearchBar, Breadcrumbs, RelatedTools, ToolCard, CategoryGrid components
- [DONE] AI Chatbot (lazy loaded, client-only), Cookie Consent banner
- [DONE] Sign in / Sign up pages, Dashboard (API keys + referral), Profile, Settings
- [DONE] Pricing page, About, Contact, Blog (list + [slug]), API Docs
- [DONE] Admin analytics page, Billing success page, Embed Generator
- [DONE] SEO: 8 sitemap files, robots.ts, OpenGraph image generators, JSON-LD structured data
- [DONE] Loading skeletons for all major routes
- [DONE] PDF Tools: 26 tools — 100% complete with dynamic [pdfTool] routing + 29 configs
- [DONE] Calculators: 21 tools — 100% complete (individual pages + client components)
- [DONE] Developer Tools: 19 tools — 100% complete
- [DONE] Image Tools: 38 tools — 100% complete
- [DONE] Text Tools: 10 tools — 100% complete
- [DONE] Unit Converters: 8 tools — 100% complete
- [DONE] Security Tools: 8 tools — 100% complete
- [DONE] Legal Tools: 8 tools — 100% complete
- [DONE] Real Estate: 8 tools — 100% complete
- [DONE] Network Tools: 7 tools — 100% complete
- [DONE] Business Tools: 6 tools — 100% complete
- [DONE] Environment Tools: 6 tools — 100% complete
- [DONE] Document Tools: 9 tools — 100% complete
- [DONE] Express API: auth routes (register/login/JWT/sessions), billing routes (Stripe), PDF processing (20 routes via Gotenberg)

[DONE] API: calculator routes, developer routes, URL shortener, temp email, SSL checker, IP checker

[DONE] API: rate limiting, CSRF protection, sanitizer, security headers, error handler middleware

[DONE] Stripe billing integration (checkout + webhook routes)

[DONE] JWT authentication system with bcrypt password hashing

[DONE] Makefile: 40+ commands covering dev, build, lint, test, database, docker, deploy

Remaining Items (TODO)

[ENG-CONFIGS] Engineering calc functions: 104/126 missing — Civil/Structural (17), Mechanical (17), HVAC (9), Chemical (9), Geotechnical (8), Water Resources (7), Electronics (7), Aerospace (6), IT Infra (9), Project Management (8)

[MED-CONFIGS] Medical calc functions: 88/113 missing — Cardiology (11), Nephrology (9), Critical Care (10), Psychiatry (8), Endocrinology (7), Pharmacology (12), OB/GYN (9), Pulmonology (6), Gastroenterology (5), Oncology (7), Pediatrics (7), Wellness (11)

[ADMIN] Admin panel: user management, content management, tool moderation, blog CMS

[EMAIL] Email templates via Resend: welcome email, password reset, email verification, newsletter

[AI-TOOLS] Wire AI-powered tools to backend: grammar checker, text summarizer, business AI tools

[IMAGE-AI] Connect image processing tools to AI service: background remover, upscaler, colorizer

[BILLING] Configure real Stripe PRICE_IDs, test full checkout + webhook flow in production

[BLOG-CMS] Build blog content pipeline / CMS for managing posts

[DOC-GEN] Complete PDF export logic for document generators (resume, invoice, contracts)

Partially Done

[STRIPE] Code complete, needs real PRICE_ID env vars and production test

[EMAIL] Resend route exists, email HTML templates not yet built

[ADMIN] Only 1 analytics page exists, no full admin dashboard

[BLOG] Structure + [slug] routing ready, no CMS or content pipeline

[AI-UI] All AI tool UIs built, backend connections incomplete

6. Priority Next Steps

1

Engineering Configs (HIGH PRIORITY)

Write 104 missing calculation functions in apps/web/src/app/tools/engineering/_components/engineering-configs.ts. Each config needs: name, description, icon, formula, reference, inputs[], outputs[], and calculate() function. Subcategories: Civil/Structural, Mechanical, HVAC, Chemical, Geotechnical, Water Resources, Electronics, Aerospace, IT Infrastructure, Project Management.

2

Medical Configs (HIGH PRIORITY)

Write 88 missing calculation functions in apps/web/src/app/tools/medical/_components/medical-configs.ts. Each config needs the same structure. Subcategories: Cardiology, Nephrology, Critical Care, Psychiatry, Endocrinology, Pharmacology, OB/GYN, Pulmonology, Gastroenterology, Oncology, Pediatrics, General Wellness.

3

Admin Panel

Build admin pages: user management (list/search/ban), tool management (DRAFT/PUBLISHED/ARCHIVED status), blog CMS (create/edit/publish posts), tool request moderation, analytics dashboard with charts.

4

Email Templates

Build Resend email HTML templates: welcome email, email verification, password reset, newsletter subscription confirmation, billing receipts. Add to API email routes.

5

AI Tool Backends

Connect grammar checker, text summarizer, text-to-speech, and business AI tools (sales copy, ad creative, SEO optimizer) to Claude API or OpenAI API in apps/api.

6

Image Processing AI

Integrate AI service for: background remover, image upscaler, image colorizer, photo-to-cartoon. Options: remove.bg API, Replicate.com, or Cloudinary AI transformations.

7

Stripe Go-Live

Set STRIPE_PRICE_PRO, STRIPE_PRICE_TEAM, STRIPE_PRICE_API with real Stripe price IDs. Test checkout flow end-to-end. Configure webhook endpoint in Stripe dashboard. Test subscription lifecycle.

8

Blog CMS

Build blog content management: markdown editor in admin, image upload to S3, publish/draft workflow, tags management, RSS feed generation.

7. Environment Variables

Copy .env.example to .env and fill in real values. Run 'make env-check' to validate.

Database

DATABASE_URL	postgresql://user:pass@localhost:5432/toolverse
DATABASE_URL_DIRECT	postgresql://user:pass@localhost:5432/toolverse
POSTGRES_USER	toolverse
POSTGRES_PASSWORD	changeme
POSTGRES_DB	toolverse

Redis

REDIS_URL	redis://localhost:6379
REDIS_PASSWORD	(empty for local)

App

NODE_ENV	development | production
PORT	4000
CORS_ORIGIN	http://localhost:3000
SITE_URL	https://roughtools.com
NEXT_PUBLIC_API_URL	http://localhost:4000
NEXT_PUBLIC_APP_URL	http://localhost:3000

Auth

JWT_SECRET	Random 256-bit secret
JWT_EXPIRES_IN	7d
BCRYPT_ROUNDS	12
CSRF_SECRET	Random secret
NEXTAUTH_SECRET	Random secret
NEXTAUTH_URL	http://localhost:3000

Email (Resend)

RESEND_API_KEY	re_xxxxxxxxxxxxxxxxxxx
EMAIL_FROM	hello@roughtools.com
SUPPORT_EMAIL	support@roughtools.com

Stripe Billing

STRIPE_SECRET_KEY	sk_test_xxx or sk_live_xxx
STRIPE_PUBLISHABLE_KEY	pk_test_xxx or pk_live_xxx
STRIPE_WEBHOOK_SECRET	whsec_xxx

STRIPE_PRICE_PRO	price_xxx (Pro plan)
STRIPE_PRICE_TEAM	price_xxx (Team plan)
STRIPE_PRICE_API	price_xxx (API plan)

Monitoring & Analytics

SENTRY_DSN	https://xxx@xxx.ingest.sentry.io/xxx
NEXT_PUBLIC_GA_MEASUREMENT_ID	G-XXXXXXXXXX
NEXT_PUBLIC_GSC_VERIFICATION_TOKEN	(Google Search Console token)
NEXT_PUBLIC_ADSENSE_CLIENT	ca-pub-XXXXXXXXXXXXXXXX
NEXT_PUBLIC_CLARITY_PROJECT_ID	XXXXXXXXXX

Storage (AWS S3)

S3_BUCKET	roughtools-uploads
AWS_ACCESS_KEY_ID	AKIAXXXXXXXXXXXXXXXX
AWS_SECRET_ACCESS_KEY	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
AWS_REGION	us-east-1

PDF Processing

GOTENBERG_URL	http://localhost:3030
ILOVEPDF_API_KEY	(optional fallback)

Internal / Security

IP_HASH_SECRET	32+ char random string
ADMIN_API_KEY	Long random string for admin API

8. Makefile Commands (40+)

Setup & Install

`make setup`	Full first-time setup (env + install + docker + db + seed)
`make install`	Install all pnpm dependencies
`make install-frozen`	Install with frozen lockfile (for CI)
`make env-setup`	Copy .env.example to .env
`make env-check`	Validate all required env vars are set

Development

`make dev`	Start web + api in parallel
`make dev-web`	Start Next.js web app only (port 3000)
`make dev-api`	Start Express API only (port 4000)
`make dev-full`	Start Docker services + all apps

Build

`make build`	Build all apps via Turborepo
`make build-web`	Build Next.js app
`make build-api`	Build Express API
`make build-packages`	Build all shared packages

Lint, Format & Types

`make lint`	Lint all apps and packages
`make lint-fix`	Auto-fix lint issues
`make format`	Format with Prettier
`make format-check`	Check formatting without writing
`make type-check`	TypeScript type checks across monorepo

Testing

`make test`	Run all tests
`make test-watch`	Run tests in watch mode
`make test-coverage`	Run tests with coverage report

Database

`make db-generate`	Generate Prisma client from schema
`make db-push`	Push schema to DB without migration (dev only)
`make db-migrate`	Create + apply new migration (prompts for name)
`make db-migrate-prod`	Apply pending migrations (production)
`make db-seed`	Seed the database

`make db-reset`	Reset + re-seed (DESTRUCTIVE — dev only)
`make db-studio`	Open Prisma Studio GUI at localhost:5555
`make db-status`	Show pending migrations

Docker

`make docker-up`	Start PostgreSQL + Redis + Gotenberg
`make docker-down`	Stop all Docker services
`make docker-logs`	Tail Docker service logs
`make docker-rebuild`	Rebuild Docker images from scratch
`make docker-ps`	Show running containers
`make docker-db`	Connect to PostgreSQL via psql
`make docker-redis`	Connect to Redis CLI

Clean

`make clean`	Remove .next + dist build artifacts
`make clean-all`	Remove node_modules + .turbo + all build output
`make clean-cache`	Clear Turborepo cache only

Deploy & Health

`make deploy-web`	Deploy web app to Vercel (production)
`make deploy-web-preview`	Deploy preview to Vercel
`make health`	Check web + API health endpoints
`make health-web`	Check http://localhost:3000/api/health
`make health-api`	Check http://localhost:4000/health

Audit & CI

`make audit`	Run pnpm security audit
`make audit-fix`	Auto-fix security vulnerabilities
`make deps-outdated`	Show outdated dependencies
`make ci`	Full CI pipeline: lint + types + test + build
`make reset`	Full project reset: clean + reinstall + setup

9. Build Output & Performance

Total Pages	707 static pages (SSG — all pre-rendered at build time)
Homepage JS	~92kB first load JavaScript
Tool Pages JS	~93–103kB first load JavaScript
Shared Chunk	87.7kB shared JS chunk
Build Type	Static Site Generation (SSG) — no server-side rendering
CDN Headers	s-maxage=86400 + stale-while-revalidate=604800 on /tools/* routes
Font	Inter only (Syne, DM_Sans, DM_Mono removed)
Lazy Loading	Chatbot + CookieConsent dynamically imported (ssr: false)
Image Opt	next/image with OptimizedImage.tsx wrapper
Code Splitting	Turborepo + Next.js automatic code splitting

Performance Optimizations Applied

- Removed unused fonts (Syne, DM_Sans, DM_Mono) — only Inter remains
- Chatbot lazy loaded (dynamic import, ssr: false) — not in initial bundle
- CookieConsent lazy loaded (dynamic import, ssr: false)
- CategoryHubLayout converted to server component (no 'use client') — reduces client JS
- Console.log stripped in production (next.config.ts compiler.removeConsole)
- CDN cache headers on /tools/* routes (s-maxage=86400, stale-while-revalidate=604800)
- Loading skeletons on all major routes (prevents layout shift)
- LazySection component uses IntersectionObserver for below-fold content
- OptimizedImage.tsx wraps next/image for consistent loading behavior
- JSON-LD structured data injected server-side (not blocking client render)

10. Social Links & Contact

Twitter/X	https://x.com/roughtools
GitHub	https://github.com/roughtools
Instagram	https://instagram.com/roughtools
LinkedIn	https://linkedin.com/company/roughtools
Email	shahidwattoo0111@gmail.com
Site	https://roughtools.com

Generated May 26, 2026 | 405 Tools | 15 Categories | ~70% Complete